

RTL TO GDS-II WORKSHOP

Project Report on
“RTL to GDS-II implementation 4-bit ALU with Add, And, Or, Xor
operations”



Submitted by:

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Problem Statement:

1. Implementation 4-bit ALU with Add, And, Or, Xor operations using Verilog.
2. Conditions: -
 - a) Do not use the Half Adder cell.
 - b) Use the C shape scenario (No. 5) for floor planning.
3. Generate GDS-II layout of design using Synopsys-flowscripts and report the following:
 - a) Design area
 - b) Clock frequency (for a positive setup slack value but less than 1)
 - c) Power consumption

Github Repository Link: https://github.com/Tirth9125/alu_4bit

RTL Development

1. RTL Source Code

Filename: alu_4bit_rtl.v

```
`timescale 1ns / 1ps

module alu_4bit (
    input clk,          // Clock added
    input [3:0] A, B,    // 4-bit inputs
    input [1:0] sel,     // select line
    output reg [3:0] Y,  // result
    output reg carry     // carry for addition
);

always @(posedge clk) begin // Changed to clocked block
    case (sel)
        2'b00: begin
            {carry, Y} <= A + B; // ADD
        end

        2'b01: begin
            Y <= A & B;          // AND
            carry <= 0;
        end

        2'b10: begin
            Y <= A | B;          // OR
            carry <= 0;
        end
    end
end
```

```
    2'b11: begin
        Y <= A ^ B;      // XOR
        carry <= 0;
    end

    default: begin
        Y <= 4'b0000;
        carry <= 0;
    end
endcase
end

endmodule
```

2. Testbench Code

Filename: alu_4bit_tb.v

```
`timescale 1ns / 1ps
```

```
module tb_alu;
```

```
    // Inputs
```

```
    reg clk;
```

```
    reg [3:0] A, B;
```

```
    reg [1:0] sel;
```

```
    // Outputs
```

```
    wire [3:0] Y;
```

```
    wire carry;
```

```
// Instantiate DUT
alu_4bit uut (
    .clk(clk),
    .A(A),
    .B(B),
    .sel(sel),
    .Y(Y),
    .carry(carry)
);

// Clock generation (10 time units period)
always #5 clk = ~clk;

// FSDB dump (same as your reference code)
initial begin
    $fsdbDumpfile("alu_4bit.fsdb");
    $fsdbDumpvars(0, tb_alu);
end

// Monitor values
initial begin
    $monitor("Time=%0t | A=%b B=%b sel=%b | Y=%b carry=%b",
        $time, A, B, sel, Y, carry);
end

// Stimulus
initial begin
```

```
// Initialize
clk = 0;
A = 0;
B = 0;
sel = 0;

// Test ADD
#10 A = 4'b0011; B = 4'b0101; sel = 2'b00;

// Test AND
#10 sel = 2'b01;

// Test OR
#10 sel = 2'b10;

// Test XOR
#10 sel = 2'b11;

// More values
#10 A = 4'b1111; B = 4'b0001; sel = 2'b00;
#10 sel = 2'b01;
#10 sel = 2'b10;
#10 sel = 2'b11;

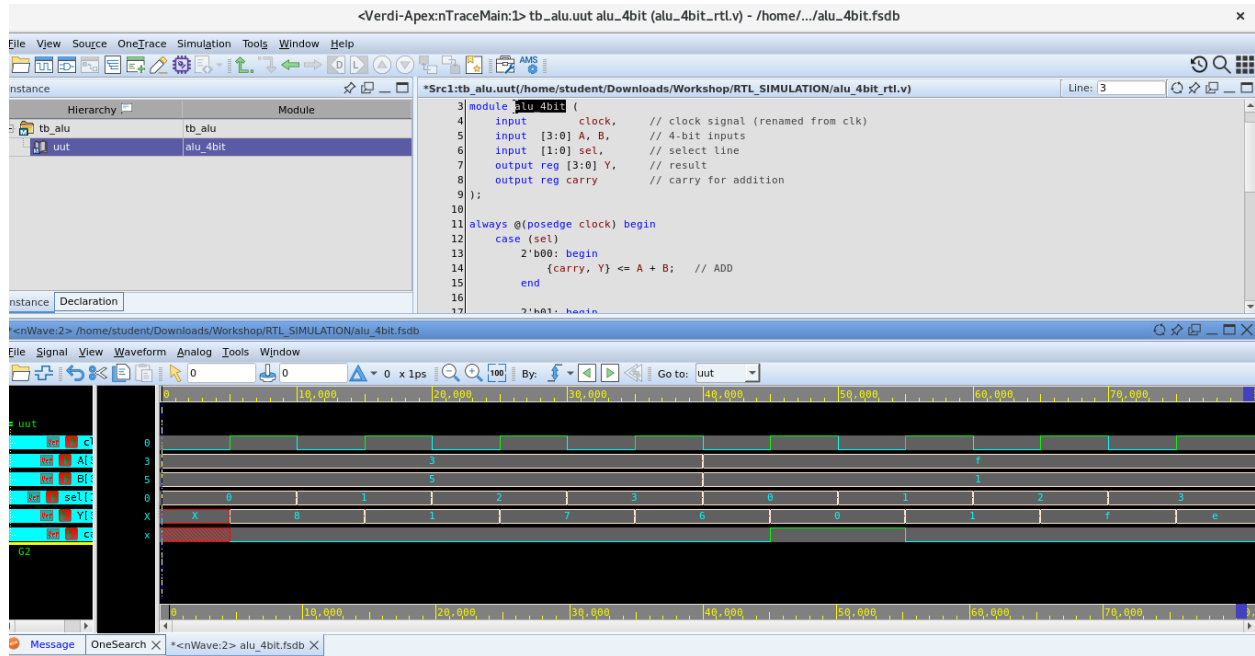
// Finish
#20 $finish;

end
```

endmodule

3. RTL Schematic:

4. Timing Diagram:



Logic Synthesis

1. Changes in common_setup.tcl File:

a) DESIGN_NAME : "alu_4bit"bit

2. Constraint File:

File name: "alu_4bit".sdc

Clock definition

```
create_clock -period 10 -name "clk" [get_ports clk]
```

```
set_clock_uncertainty 0.2 [get_clocks clk]
```

```
set_clock_transition 0.1 [get_clocks clk]
```

Input/Output delays

```
set_input_delay 2.0 -clock clk [all_inputs]
```

```
set_output_delay 2.0 -clock clk [all_outputs]
```

False paths

```
# set_false_path -from [get_ports rst_n]
```

3. Design Compiler TCL commands:

Filename: dc_script.tcl

```
#=====
```

```
# Synopsys Design Compiler
```

```
#=====
```

```
#-----
```

```
# 1. SETUP & ENVIRONMENT
```

```
#-----
```



```
set DESIGN_NAME    "alu_4bit"
set RTL_DIR        "../RTL/"
set CONSTRAINTS_FILE "../constraints/${DESIGN_NAME}.sdc"
set LIBRARY_DIR     "../ref/lib/stdcell_rvt/"
set OUTPUT_DIR      "../outputs"
set REPORTS_DIR     "../reports"
```

```
# Target library (technology .db files)
```

```
set target_library "$LIBRARY_DIR/saed32rvt_ss0p7vn40c.db"
set link_library   "* $LIBRARY_DIR/saed32rvt_ss0p7vn40c.db
$LIBRARY_DIR/saed32rvt_ff1p16v125c.db"
```

```
# Create output directories
```

```
file mkdir $OUTPUT_DIR
```

```
file mkdir $REPORTS_DIR
```

```
#-----
```

```
# 2. READ RTL SOURCE FILES
```

```
#-----
```

```
# Option A: Read Verilog
```

```
read_verilog [list \
    $RTL_DIR/${DESIGN_NAME}_rtl.v \
]
```

```
# Option B: Read SystemVerilog
```

```
# read_file -format sverilog [list \
#    $RTL_DIR/top_module.sv \
```

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```
# $RTL_DIR/submodule_a.sv \  
# ]
```

Option C: Read VHDL

```
# read_vhdl [list \  
# $RTL_DIR/top_module.vhd \  
# ]
```

Set the top-level design

current_design \$DESIGN_NAME

#-----

3. LINK THE DESIGN

#-----

link

#-----

4. SET DON'T USE CELLS

#-----

```
#set_dont_use [get_lib_cells */FADD*]
```

```
#set_dont_use [get_lib_cells */HADD*]
```

```
#set_dont_use [get_lib_cells */AO*]
```

```
#set_dont_use [get_lib_cells */OA*]
```

```
#set_dont_use [get_lib_cells */NAND*]
```

```
#set_dont_use [get_lib_cells */XOR*]
```

```
set_dont_use [get_lib_cells */NOR*]
```

```
#set_dont_use [get_lib_cells */XNOR*]
```

```
#set_dont_use [get_lib_cells */MUX*]
```

```
#-----
```

```
# 5. APPLY CONSTRAINTS (SDC)
```

```
#-----
```

```
source $CONSTRAINTS_FILE
```

```
#-----
```

```
# 6. COMPILE (SYNTHESIS)
```

```
#-----
```

```
# Standard compile
```

```
# compile -map_effort medium -area_effort medium
```

```
# Ultra compile
```

```
compile_ultra
```

```
# Incremental compile for further optimization (optional)
```

```
# compile_ultra -incremental
```

```
# Fix hold violations
```

```
#set_fix_hold [all_clocks]
```

```
#compile_ultra -incremental -only_hold_time
```

```
#-----
```

```
# 7. POST-COMPILE OPTIMIZATIONS
```

#-----

Remove unconnected ports/nets

remove_unconnected_ports [get_cells -hierarchical *]

#-----

8. REPORTS

#-----

Timing reports

report_timing -path full -delay max -max_paths 10 \
> \$REPORTS_DIR/timing_setup.rpt

report_timing -path full -delay min -max_paths 10 \
> \$REPORTS_DIR/timing_hold.rpt

Area report

report_area -hierarchy \
> \$REPORTS_DIR/area.rpt

Power report

report_power -hierarchy \
> \$REPORTS_DIR/power.rpt

Design rule checks

report_constraint -all_violators \
> \$REPORTS_DIR/constraints.rpt

```
report_cell \  
    > $REPORTS_DIR/cells.rpt
```

```
report_net \  
    > $REPORTS_DIR/nets.rpt
```

```
# QoR summary
```

```
report_qor \  
    > $REPORTS_DIR/qor.rpt
```

```
#-----  
# 9. WRITE OUTPUTS  
#-----
```

```
# Remove multi-port nets and constant logic assign statements from netlist  
#set_fix_multiple_port_nets -all -buffer_constants
```

```
# Write gate-level netlist (Verilog)  
write -format verilog -hierarchy \  
    -output $OUTPUT_DIR/${DESIGN_NAME}_netlist.v
```

```
# Write SDC constraints  
write_sdc $OUTPUT_DIR/${DESIGN_NAME}.sdc
```

```
# 10. DONE
```

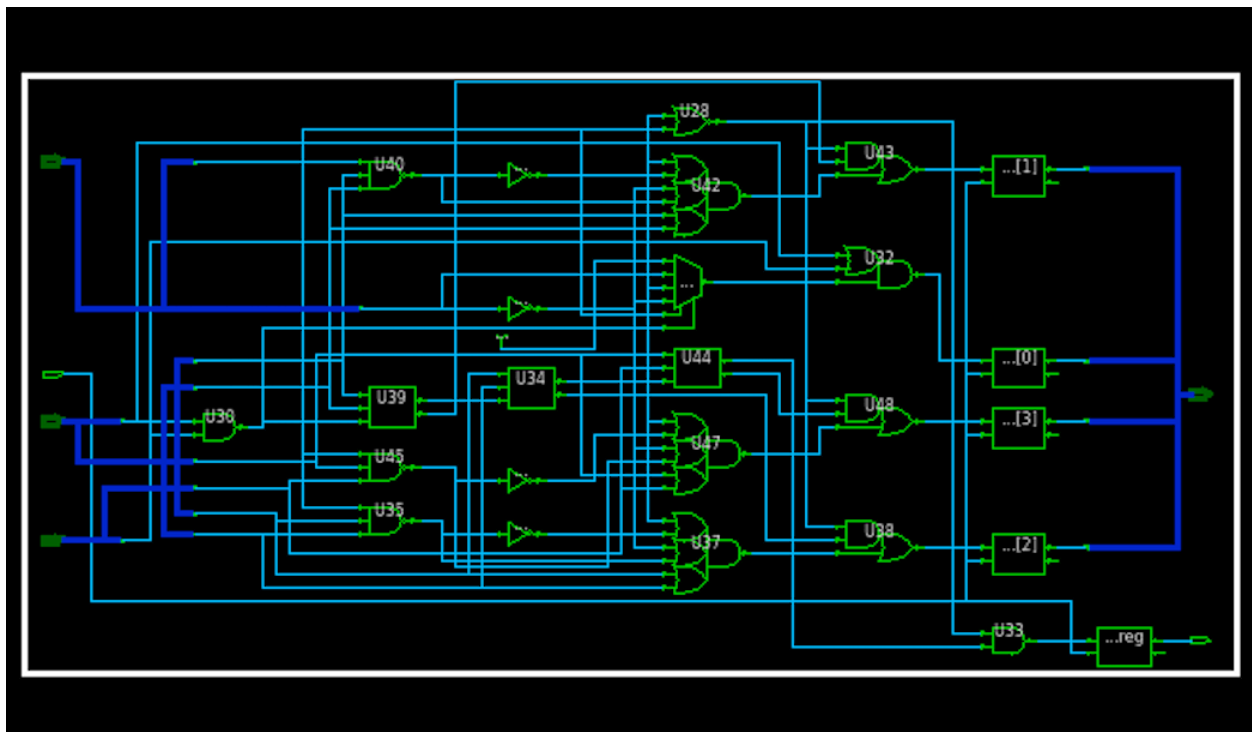
```
echo "=====  
echo " Synthesis COMPLETE: $DESIGN_NAME"
```

Parameters changed: -

- RTL_SOURCE_FILES : “./../rtl_simulation/alu_4bit.v”
- Half Adder cells set to don't use [To increase the number of cells in design]

4. Generated outputs:

1. Schematics:



2. Report content:

```

Activities Terminal Apr 5 13:11
student@ict-chipin:~/Downloads/Workshop/DC

File Edit View Search Terminal Help
license file(s): 27020@c2s.cdacb.in
[SCL] 04/05/2026 13:08:46 PID:23929 Client:ict-chipin.sot.pdpu.ac.in Authorization failed DesignWare-Foundation-Ultra 2024.09
Information: Evaluating DesignWare library utilization. (UISN-27)

=====
| DesignWare Building Block Library | Version | Available |
=====
| Basic DW Building Blocks | W-2024.09-DWBB_202409.0 | * |
| Licensed DW Building Blocks | W-2024.09-DWBB_202409.0 | * |
=====

=====
| Flow Information |
-----
| Flow | Design Compiler WLM |
| Command Line | compile_ultra |
=====

=====
| Design Information | Value |
=====
| Number of Scenarios | 0 |
| Leaf Cell Count | 68 |
| Number of User Hierarchies | 0 |
| Sequential Cell Count | 11 |
| Macro Count | 0 |
| Number of Power Domains | 0 |
| Number of Path Groups | 2 |
| Number of VT Class | 1 |
| Number of Clocks | 1 |
| Number of Dont Touch Cells | 18 |
| Number of Dont Touch Nets | 0 |
| Number of Size Only Cells | 0 |
| Design with UPF Data | false |
=====

Information: Sequential output inversion is enabled. SVF file must be used for formal verification. (OPT-1208)
Simplifying Design 'pow2_3bit'

```

Category	Parameter	Value
Timing (Hold)	Worst Hold Slack	0.74 ns
Timing (Setup)	Worst Setup Slack	6.55 ns
Area	Total Cells	24
	Combinational Cells	10
	Sequential Cells	11
	Buffers/Inverters	1
	Total Cell Area	96.83 μm^2
	Total Area	103.34 μm^2
Power	Switching Power	0.0645 μW
	Internal Power	3.272 μW
	Leakage Power	0.155 μW
	Total Power	3.492 μW

Physical Design

1. Floor-planning:

#scenario5:

```
initialize_floorplan -shape L -control_type die -side_length {20 30 20 20} -core_offset {5}
```

```
set_individual_pin_constraints -ports [get_ports clock] -exclude_sides {1 2 3 4 5}
```

```
place_pins -self
```

```
create_placement -floorplan -effort low
```

Output pins (y[7:0])

```
set_individual_pin_constraints \
```

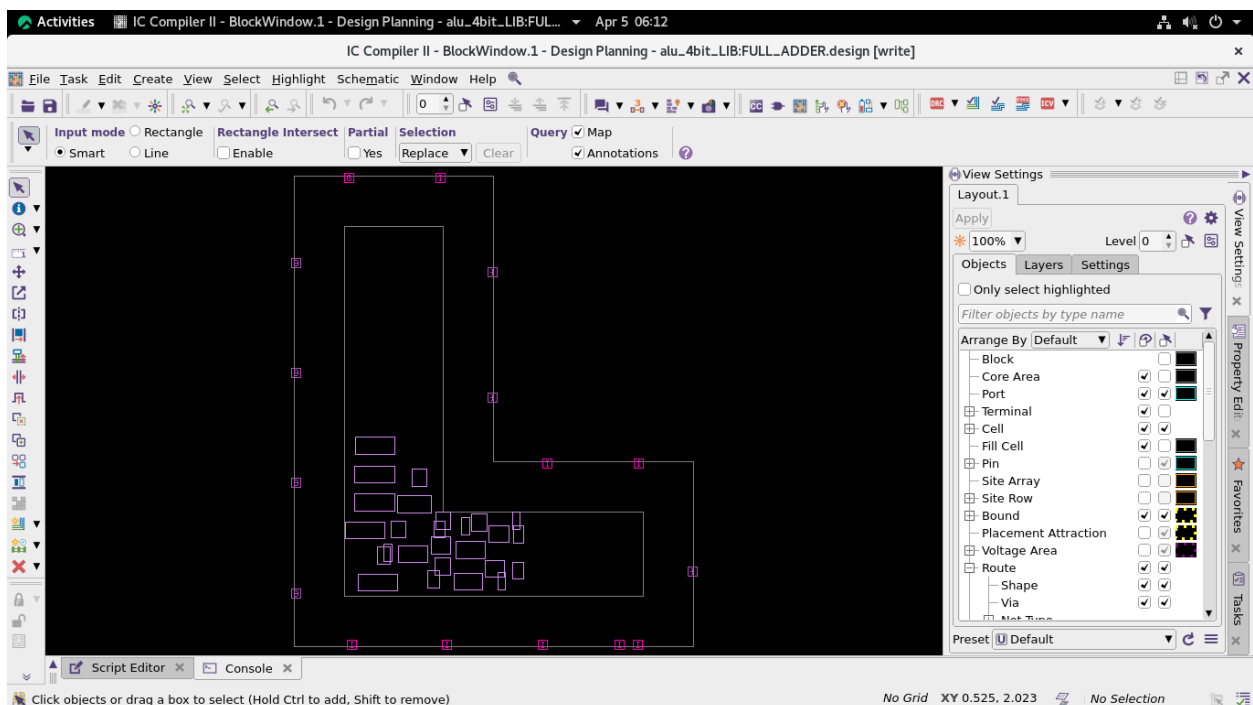
```
-ports [get_ports {y[*]}] \
```

```
-sides {3 4} \
```

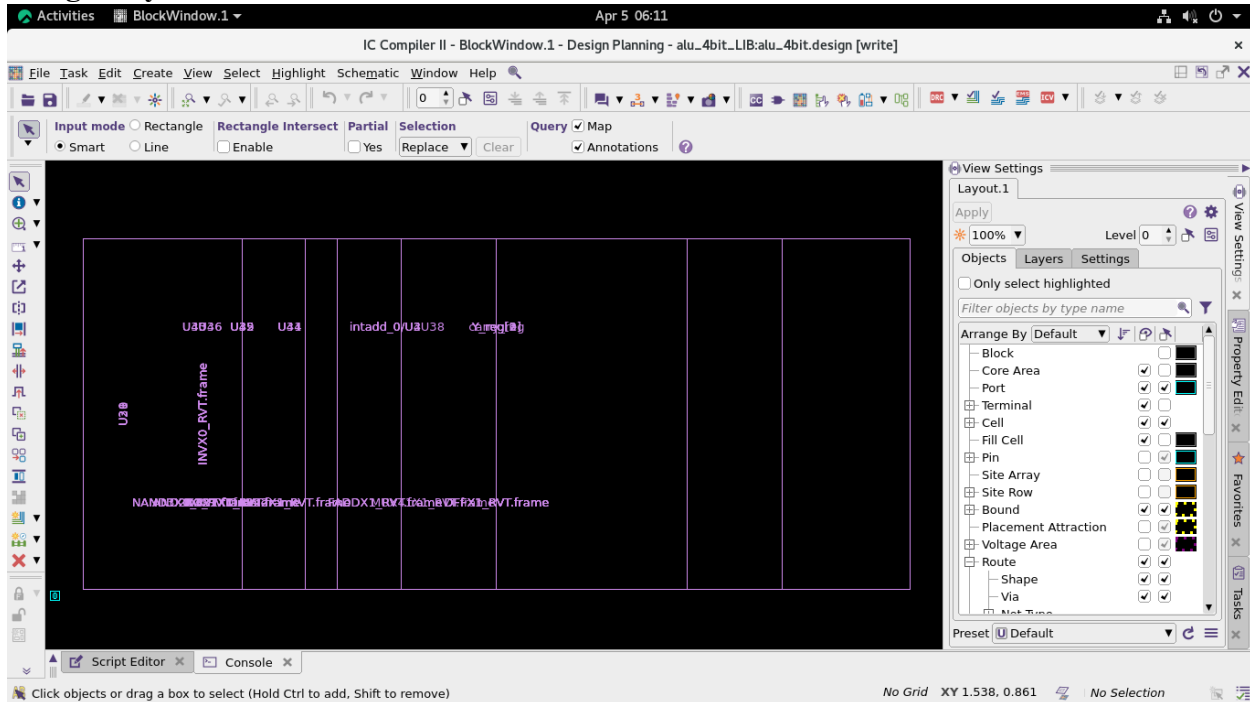
```
-pin_spacing_distance 2
```

```
place_pins -self
```

```
create_placement -floorplan -effort medium
```



Design Layout:



2. Power planning:

#-----

4b. Power / ground rings

#-----

```
create_pg_ring_pattern core_ring_pattern -horizontal_layer M7 -horizontal_width .4 -
horizontal_spacing .3 -vertical_layer M8 -vertical_width .4 -vertical_spacing .3
```

```
set_pg_strategy core_power_ring -core -pattern {{name : core_ring_pattern}} {nets : {VDD
VSS}} {offset : {.5}}}}
```

```
compile_pg -strategies core_power_ring
```

#-----

4c. Power mesh

#-----

```
create_pg_mesh_pattern mesh -layers { {{vertical_layer: M6}} {width: .34} {spacing:
interleaving} {pitch: 5} {offset: .5}} {{horizontal_layer: M7}} {width: .38} {spacing:
```

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```
interleaving} {pitch: 5} {offset: .5}} {{vertical_layer: M8} {width: .38} {spacing:  
interleaving} {pitch: 5} {offset: .5}}}
```

```
set_pg_strategy core_mesh -pattern { {pattern:mesh} {nets: VDD VSS}} -core -extension {stop:  
innermost_ring}
```

```
compile_pg -strategies core_mesh
```

```
#-----
```

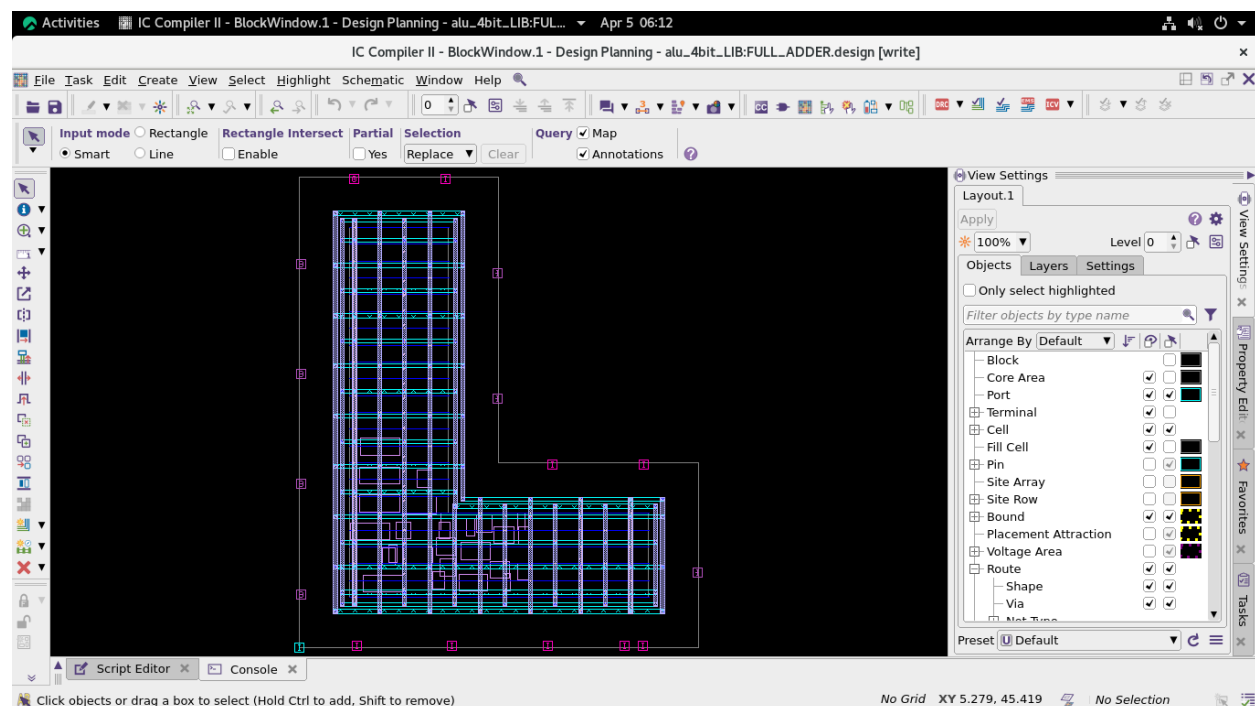
```
# 4d. Standard-cell power rails
```

```
#-----
```

```
create_pg_std_cell_conn_pattern std_cell_rail -layers {M1} -rail_width 0.06
```

```
set_pg_strategy rail_strat -core -pattern {{name: std_cell_rail} {nets: VDD VSS} }
```

```
compile_pg -strategies rail_strat
```



3. Placement:

Set NOR cells to don't use [Similar constraint during Logic Synthesis].

```
#set_dont_use [get_lib_cells */FADD*]
```

```
#set_dont_use [get_lib_cells */HADD*]
```

```
#set_dont_use [get_lib_cells */AO*]
```

```
#set_dont_use [get_lib_cells */OA*]
```

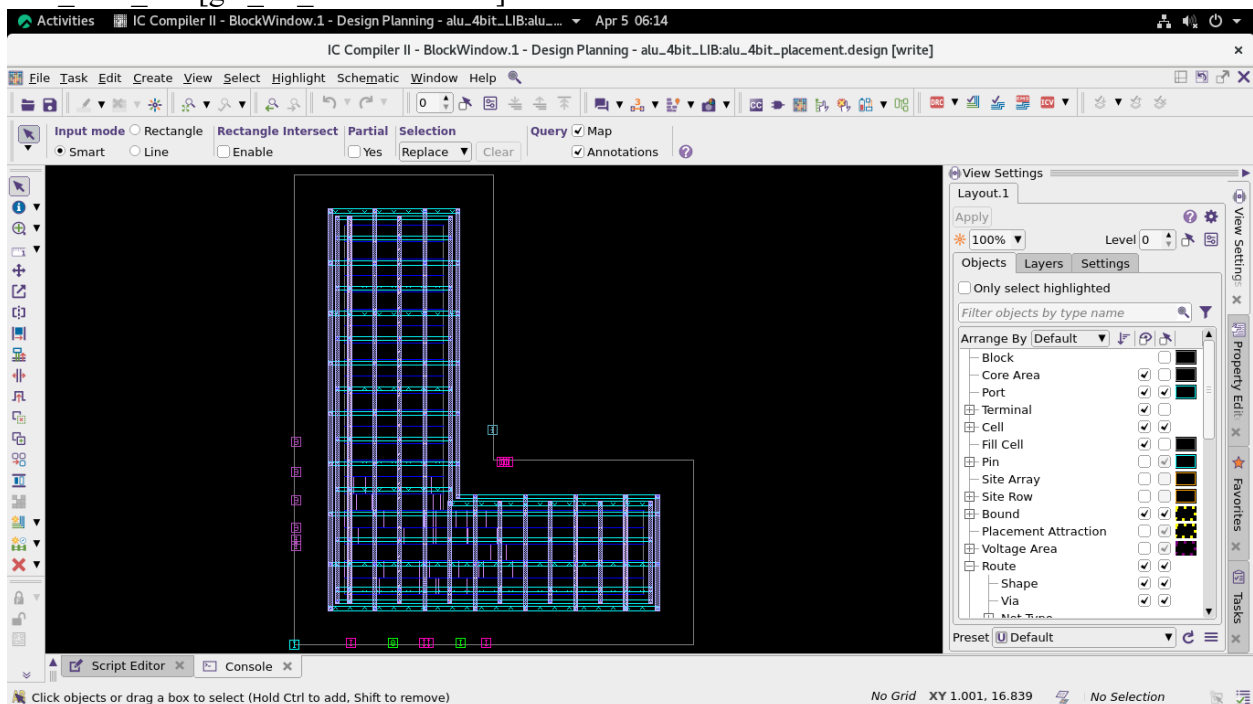
```
#set_dont_use [get_lib_cells */NAND*]
```

```
#set_dont_use [get_lib_cells */XOR*]
```

```
set_dont_use [get_lib_cells */NOR*]
```

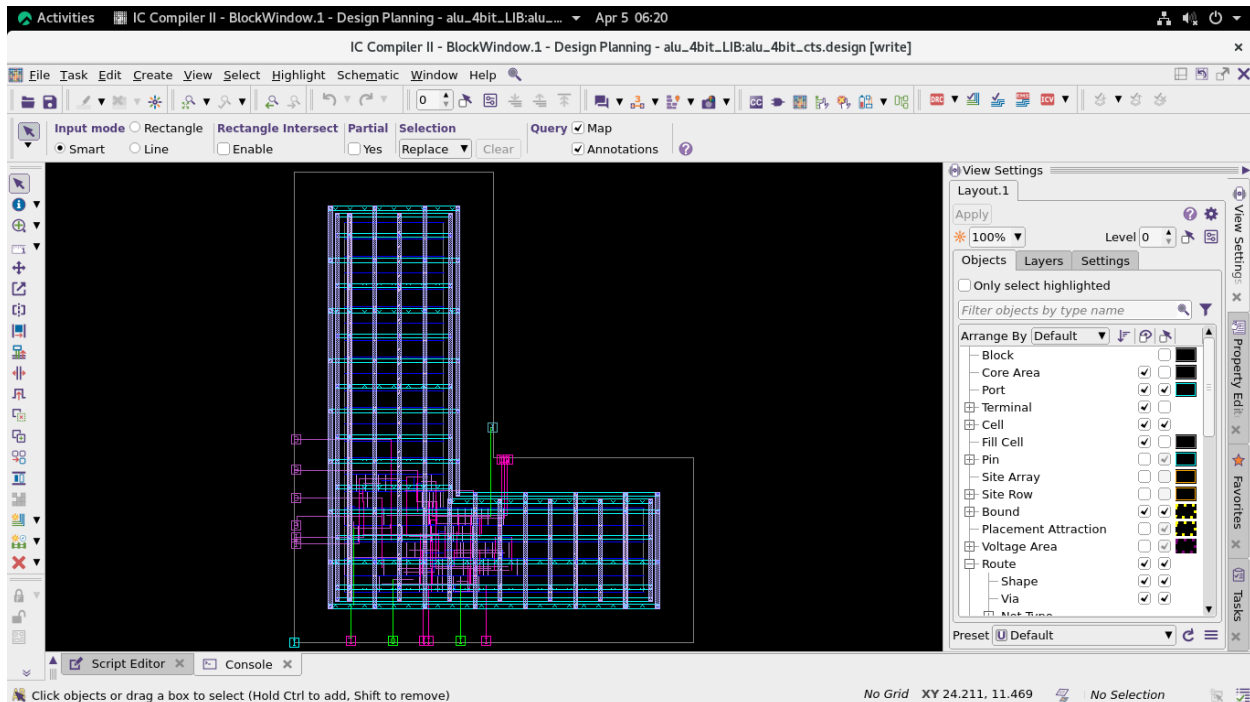
```
#set_dont_use [get_lib_cells */XNOR*]
```

```
#set_dont_use [get_lib_cells */MUX*]
```



4. Clock Tree synthesis:

Design layout:



5. Routing:

7a. Global route options

```
set_app_options -name route.global.timing_driven -value true
```

```
set_app_options -name route.global.crosstalk_driven -value false
```

```
#-----
```

7b. Track assignment options

```
set_app_options -name route.track.timing_driven -value true
```

```
set_app_options -name route.track.crosstalk_driven -value true
```

```
#-----
```

7c. Detail route options

```
set_app_options -name route.detail.timing_driven -value true
```

```
#set_app_options -name route.detail.save_after_iterations -value false
```

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```
set_app_options -name route.detail.force_max_number_iterations -value false
```

```
set_app_options -name route.detail.antenna -value true
```

```
set_app_options -name route.detail.antenna_fixing_preference -value use_diodes
```

```
set_app_options -name route.detail.diode_libcell_names -value */ANTENNA_RVT
```

```
#-----
```

```
# 7d. Atomic routing sequence
```

```
route_global
```

```
route_track
```

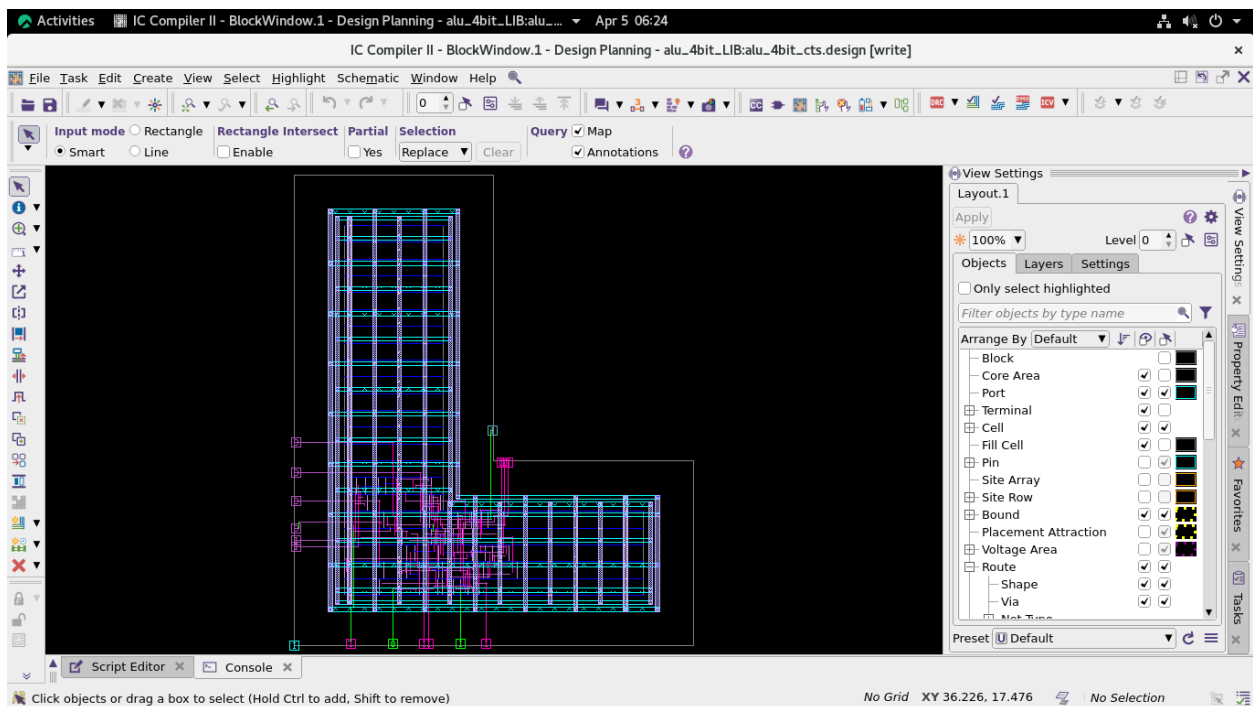
```
route_detail
```

```
#-----
```

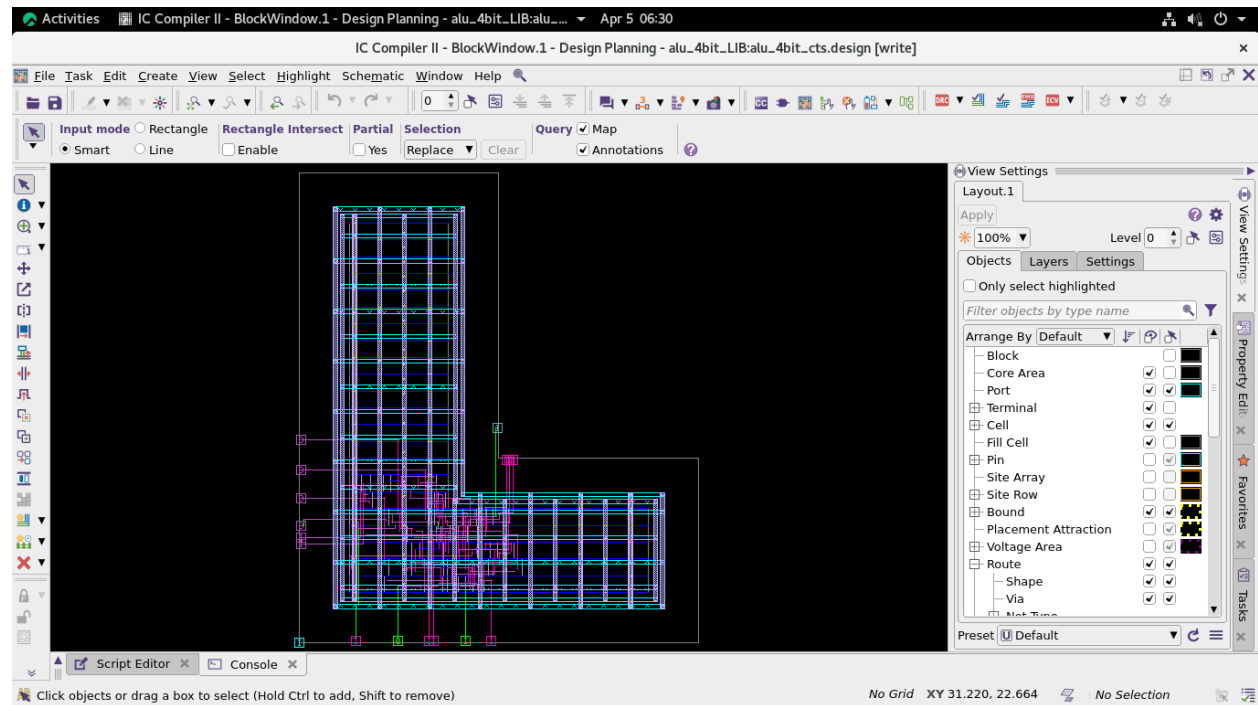
```
# 7e. Post-route optimisation
```

```
#-----
```

```
route_opt
```



Final IC Layout after all the Physical design flow:



```
Date : Sun Apr 5 15:21:53 2026
*****
No mismatches exist on the design.
Information: Fill shapes with 'active_fill' purpose will be skipped from output DEF. (DEFW-032)
-----
Number of Written DEF Constructs
-----
VERSION : 1
DIVIDERCHAR : 1
BUSBITCHARS : 1
DESIGN : 1
UNITS : 1
PROPERTYDEFINITIONS : 1
DIEAREA : 1
ROW : 9
TRACKS : 39
VIAS : 7
COMPONENTS : 29
PINS : 15
PINPROPERTIES : 13
SPECIALNETS : 28
NETS : 38
[icc2-lic Sun Apr 5 15:21:53 2026] Command 'write parasitics' requires licenses
Information: The command 'write parasitics' cleared the undo history. (UNDO-016)
Information: Design pow2_3bit has 39 nets, 0 global routed, 37 detail routed. (NEX-024)
NEX: extract design pow2_3bit
Information: batch extract takes 0.00 seconds, elapse 0.00 seconds (NEX-015)
Information: The stitching and editing of coupling caps is turned OFF for design 'FULL_ADDER_LIB:pow2_3bit.design'. (TIM-125)
NEX: write corner ID 0 parasitics to ./outputs/pow2_3bit_final.spef.p1_125.spef, corner name nom
spefName ./outputs/pow2_3bit_final.spef.p1_125.spef, corner name nom
NEX: write corner ID 1 parasitics to ./outputs/pow2_3bit_final.spef.p2_125.spef
spefName ./outputs/pow2_3bit_final.spef.p2_125.spef, corner name nom
NEX:Warning write_parasitics supports only Ohm and kOhm as user resistance unit not 1M0hm, we force to set it Ohm.NEX: write_parasitics command finished

Warning: Multiply cell derates are skipped!

Warning: Multiply net derates are skipped!
Information: Saving block 'FULL_ADDER_LIB:pow2_3bit.design'
Saving library 'FULL_ADDER_LIB'
=====
ICC2 PnR COMPLETE: pow2_3bit
=====
```

Post Physical Design report file analysis**1. Floorplanning:**

Slack	0.74 ns
Power consumption	3.492 μ W
Area	103.34 μ m ²

2. Powerplanning and Placement:

Slack	0.74 ns
Power consumption	3.492 μ W
Area	103.34 μ m ²
Connectivity	Fully connected

3. Clock Tree Synthesis:

Slack	0.74 ns
Power consumption	3.492 μ W
Area	103.34 μ m ²
Clock latency	~0.07 ns (approx.)

4. Routing:

Slack	0.74 ns
Power consumption	3.492 μ W
Area	103.34 μ m ²
PG Connectivity	Fully connected
Routing Verification	No open nets

Static Timing Analysis

1. Late corner:

```
set report_default_significant_digits 6
set link_path "../ref/lib/stdcell_rvt/saed32rvt_ff1p16v125c.db"

set DESIGN_NAME "alu_4bit"
set REPORTS_DIR "./reports"
file mkdir $REPORTS_DIR

read_verilog "../ICCII/outputs/${DESIGN_NAME}.routed.v"
link_design
current_design $DESIGN_NAME

read_sdc "../ICCII/outputs/${DESIGN_NAME}_final.sdc"

# Late corner SPEF (slow)
read_parasitics "../ICCII/outputs/${DESIGN_NAME}_func::nom.spef.p1_125.spef"

update_timing -full

report_timing -path full -delay max -max_paths 10 \
    > $REPORTS_DIR/timing_setup.rpt

report_design > $REPORTS_DIR/design_setup.rpt
```

```
*****
Report : timing
        -path_type full
        -delay_type min
        -slack_lesser_than 0.000000
        -max_paths 10
        -sort_by slack
Design : pow2_3bit
Version: W-2024.09
Date   : Sun Apr  5 15:38:53 2026
*****

No paths with slack less than 0.000000.
```

2. Early corner:

```
set report_default_significant_digits 6
set link_path "../ref/lib/stdcell_rvt/saed32rvt_fflp16v125c.db"

set DESIGN_NAME "alu_4bit"
set REPORTS_DIR "./reports"
file mkdir $REPORTS_DIR

read_verilog "../ICCII/outputs/${DESIGN_NAME}.routed.v"
link_design
current_design $DESIGN_NAME

read_sdc "../ICCII/outputs/${DESIGN_NAME}_final.sdc"

# Early corner SPEF (fast)
read_parasitics "../ICCII/outputs/${DESIGN_NAME}_func::nom.spef.p2_125.spef"
```

```
update_timing -full
```

```
report_timing -path full -delay min -max_paths 10 \  
> $REPORTS_DIR/timing_hold.rpt
```

```
report_design > $REPORTS_DIR/design_hold.rpt
```

```
*****  
Report : timing  
        -path_type full  
        -delay_type max  
        -slack_lesser_than 0.000000  
        -max_paths 10  
        -sort_by slack  
Design : pow2_3bit  
Version: W-2024.09  
Date   : Sun Apr  5 15:38:53 2026  
*****  
  
No paths with slack less than 0.000000.
```

3. Report Content:

Slack	6.55 ns (setup), 0.74 ns (hold)
Violations	None
Timing Status	MET

Final Results

Cell Area	96.83 μm^2
Total Area	103.34 μm^2
Power Consumption	3.492 μW
Clock Frequency	~100 MHz (10 ns clock)

Learnings

I. RTL Development:

- To implement a circuit using Behavioural modelling in Verilog HDL
- To simulate a Verilog testbench using VCS and Verdi tools

II. Logic Synthesis

- Using the Design Compiler tool to build the gate-level logic design of the provided Verilog module • Setting up the constraints and updating constraints to adjust the outputs
- Understood the concepts like worst setup, slack, and process corners, which create an impact on our design

III. Physical Design.

- Learned to create a floorplan and power plan for the physical design using the IC Compiler tool, and to place cells based on the power plan
- Learned to check whether the design is connected or not to verify the above steps
- Learned to synthesize a clock tree and check its latency
- Learned the routing process to connect each path physically in the design

IV. Static Timing Analysis

- Using the Primetime tool to perform the Static Timing Analysis of the design
- Learned to interpret the reports generated by tools for the analysis of design
- Learned to implement the VLSI Design Flow from RTL Development to STA with the help of Synopsys Tools